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Automated Vehicle Parking Slot Detection System Using Deep Learning

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***Abstract*—Traffic congestion at the parking slots is a major problem that the modern society is facing nowadays, as the vehicle numbers are increasing at a rapid pace without the increment of the parking slots. The research done here helps solve the traffic congestion problem at the bottleneck of the networks mainly at the parking slots, by Instance Segmentation algorithms and Deep Learning. The model gets all the initial available parking slots that are available in the given area and real time processing is done on the obtained data to find whether the slots are empty or occupied with any vehicle and gives the information of empty slots. Apart from locating a free parking space for a car, the model also finds out appropriate parking space for two wheelers (less space occupant vehicles). The proposed system shows improved robustness achieving a mask rate of recognition greater than 92.33% and a boundary recognition rate of 98.4%.**

Keywords—Mask R-CNN, Instance Segmentation, Vehicle Parking System, Automatic Vehicle Parking Detection, Vision Based Vehicle Parking System

I. INTRODUCTION

Traffic congestion is the major problem the modern society is facing nowadays, as the number of vehicles and their sizes are increasing day by day without the increment of the parking slots [1]. As the urbanization is increasing day by day and the number of vehicles adding to the cities are increasing with the limited number of parking slots in the cities which increasing extra fuel burn out at the time of searching for the parking slots as the person has to go and search whether a slot is empty of occupied and the engine keeps on running till the time.

According to the reports presented in [2], smart parking could save considerable amount of fuel if it is implemented properly and used according to the planning of cities and urban areas which are densely populated. In the city of Pune only, around 2.3 lakhs vehicles[3] will be added every year and it is the first urban city which has the number of vehicles more than the population of the city and the city has only around 5000 registered parking slots which shows the demand for parking slots in that place. The model gets all the parking slots that are available in the given area and real time processing is done on the obtained data to find whether the slots are empty or occupied with any vehicle and gives the information of empty slots thus, it solves the problem and reduces the traffic congestion and fuel emissions. A person has to go around and has to check all the slots to find empty parking slots. Better management and usage of intelligent parking systems will be useful to solve the problem. One study showed that about 86% of people find it difficult to find a parking space in multilevel parking lots.

II. RELATED WORK

A method using ultrasonic detector to identify the parking slots with the help of specialized LEDs [1]. It needs a large number of LEDs and sensors for detection of slots to find they are empty or occupied. The main limitation was that the sensors had to withstand climatic changes.

The installation of sensors is a difficult process. In the method used in [2], it detects the parking slots and their coordinates with the help of web cameras in all the directions of the parking. It uses OpenCV for training and the result is shown whether the slot is empty or occupied. In [3], a method incorporating RFID was put forth to Check-in and check-out are done automatically with the help of radio waves.

The whole system is connected to the internet which has a database called RFIDATA which stores all the data coming from the parking slots. If the parking space is completely filled, the barrier will be completely closed and it saves time and fuel. The main limitation of this method is setting and maintaining the hardware.

An algorithm PSDL which is a learning-based parking slot detection algorithm is used in [4]. With the help of PSDL the marking points are detected first and the type of parking slot is detected. Free space in between two vehicles, the coordinates and the type of slots are found with the help of sensors. The main drawback of this method is that it is not completely automatic as it will start to work if a car is kept in the slot by the driver manually.

III. Proposed Parking Model and Methodology

The proposed methodology uses the Artificial Neural Networks to provide a better solution for the traffic congestions [21].

Here in the parking bays during the setup of the proposed method. First, we have to segment the parking slot manually based on the live video stream obtained from the surveillance camera and each parking slot will be given an Id by the system and then each slot will be geo-tagged and mapped alongside its Id.

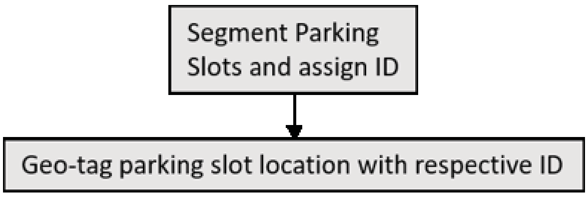

Fig. 1. Initial Setup.

After the initial setup of the system, the live video stream is obtained from the installed camera and performed a frame extraction at regular intervals, these frames are fed to the system where the free and the reserved parking slots are identified. This identified vector is sent to a database of the parking slots where all the data about the parking slots is stored.

The model that we are using to classify the vacant and booked parking slot is Mask R-CNN, this is similar to Faster R-CNN with a layer of FCN that Is used to segment the bounding box and identify the exact mask of the object in the bounding box.

The mask here is used to identify the exact area covered by the vehicle so that if the area covered by it is less than 50\%, this is reserved as partially for the sake of light motor vehicles or bikes.

This accurate area calculation also helps in accommodating any light motor vehicles in the parking spaces if a car is wrongly parked between two parking slots. This method also solves the problem of multi-camera network in parking in a bay without any duplications.

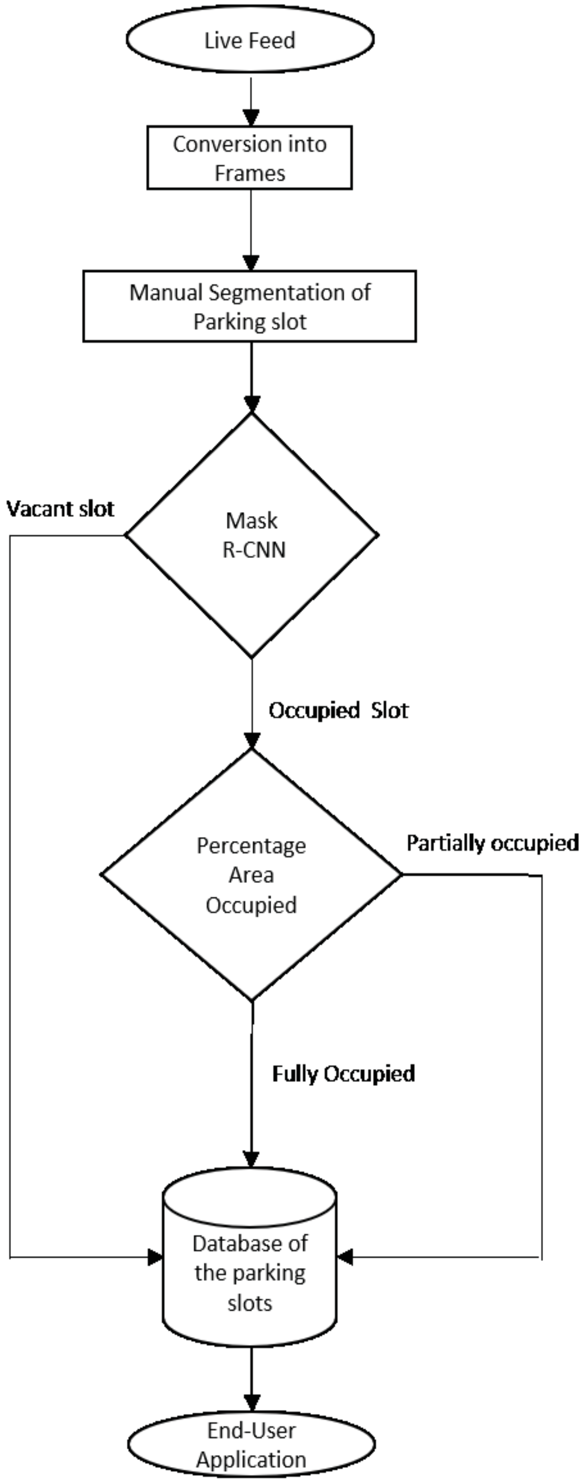

Fig. 2. Flowchart

IV. Algorithm of Proposed Methodology

1. Getting the live stream from multiple surveillance cameras.

2. Extracting frames from the live according to the preset frequency.
3. Segment each parking slot manually and assign it with an ID and Geo tag them with the same.
4. Applying mask R CNN model on the extracted frames to predict the result.
5. Area occupied by the vehicle in the parking Slot is calculated, Percentage area occupied can be calculated by the below formula.

$$\text{Area Occupied} = \frac{\text{Area Occupied by Car}}{\text{Area Occupied by the Parking Slot}} \quad (1)$$

6. If the percentage area occupied by the vehicle is more than 50% then the slot is classified as completely occupied or else it's marked partially occupied.
7. Partially occupied parking slot can be allotted with a light motor vehicle.
8. Based on the starter parking Slot is divided into three classes namely fully reserved partially reserved and unreserved.
9. All this data is sent into the database of the parking lot.
10. Now user can Reserve the parking slot according to the status of it which has been stored in the database and be directed towards it.

V. Experimental Analysis

A. Dataset

PKLOT public dataset [2] to train as well as test our proposed methodology. It is a huge public datasets with 12,417.01 parking area images and 695,899.01 parking slot images in total.

The image acquisition is done in a 5-minute time lapse and for 30 days. It has images acquired in the climatic conditions sunny, cloudy and rainy. The dataset is balanced which has 43.48% occupied parking slots and 56.42% empty slots which makes the classifier unbiased.

We are using 10,209 images for training the model and 2,208 images for testing and validating the model. The model is made robust by training it over all the weather conditions, so that it will be able to detect the cars even in the noisy environments we have annotated the images with the help of VGG Image Annotator HTML version and fitted the vehicles in a polygon so that it can be trained using our Mask R-CNN model. Fig.3. shows the Snapshot of a dataset.

Fig. 3. Snapshot of the Dataset

B. Building the Mask R-CNN Model

The objective of the task is to build a system model which works based on Mask R-CNN [13] which is able to detect whether the parking slot is empty or occupied along with the area occupied by the vehicle accurately. The first step for the process is to get the data available for the model for working. We are using PKLot [2] public dataset to check whether the model is giving the correct results or not with good accuracy.

Mask R-CNN is a segmentation model based on Neural Network Architecture using Deep learning [20] which segments at each instance and finds the location of pixels of the class. It segments different objects which are present in an image irrespective of type of objects they are and training them on the PKLot dataset which we are using for the project. There are stepwise procedures to be followed to obtain better results of Mask R-CNN. First, we annotate the collected data and form a single json file for all the dataset. Then we use transfer learning to obtain higher accuracy and in a reduced training time, we use the trained weights of the model over COCO dataset and use them as the initial weights in our training. Then we apply the Mask R-CNN model to the dataset and with the selected weights as initial weights. Two classes used are 'Background' and 'Vehicle'. The model marks the vehicle and leaves the background untouched and calculates the percentage area occupied by the vehicle in a parking bay.

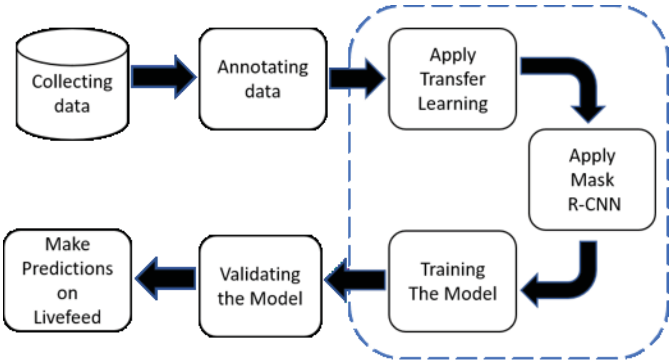

Fig. 4. Overview of Proposed Methodology

The model marks it as fully occupied if the vehicle percentage area occupied is greater than 50%, else the model detects it as a partially parked slot and allows it to be accommodated with any light motor vehicle. Then we validated our model with 2000 images that are in the test set.

C. Architecture of the Model

Mask R-CNN is used in solving instance segmentation problems in deep learning. Given an image as an input, it gives the objects with their bounding box, classes and a mask around the object. In this, the input is a CNN Feature Map and the output is a matrix with 1's on all locations where the pixel belongs to the object and 0's elsewhere.

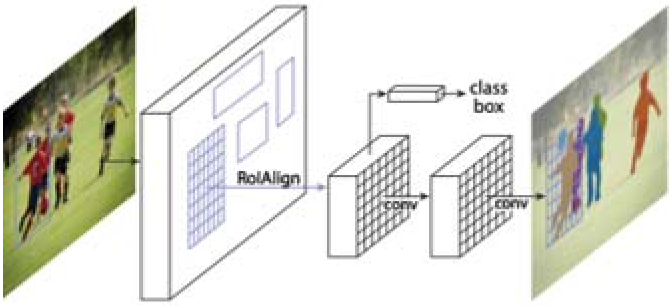

Fig. 5. Snapshots of the model's predictions

Mask R-CNN is a two-step procedure. It has a similar first stage as Faster R-CNN. The second stage ROI's binary mask was generated using the Mask R-CNN outputs a along with predicating class and box offset. The processes of bounding box and regression are done in parallel along with the classification depending on the predictions of the mask. The multi task loss during training on each sample is defined as (2)

Mask R-CNN, an extended version of Faster R-CNN [19], classifies as depicted in Fig.7. The mask branch enables a rapid experimentation and a fast system by adding only a small computational overhead.

$$L = L(class) + L(box) + L(mask) \tag{2}$$

where L(class) is the classification loss and L(box) is bounding box loss and L(mask) is the average binary entropy loss for K classes.

VI. Results and Discussion

In the car parking management system, which is used to detect vacant and occupied parking slots, one factor that has to be considered is the cost required. In case of detecting these slots with the help of sensors, the cost of implementation is very high. The preparation and maintenance of the infrastructure is also a major concern in such systems. This paper uses the concept of car parking based on vision.

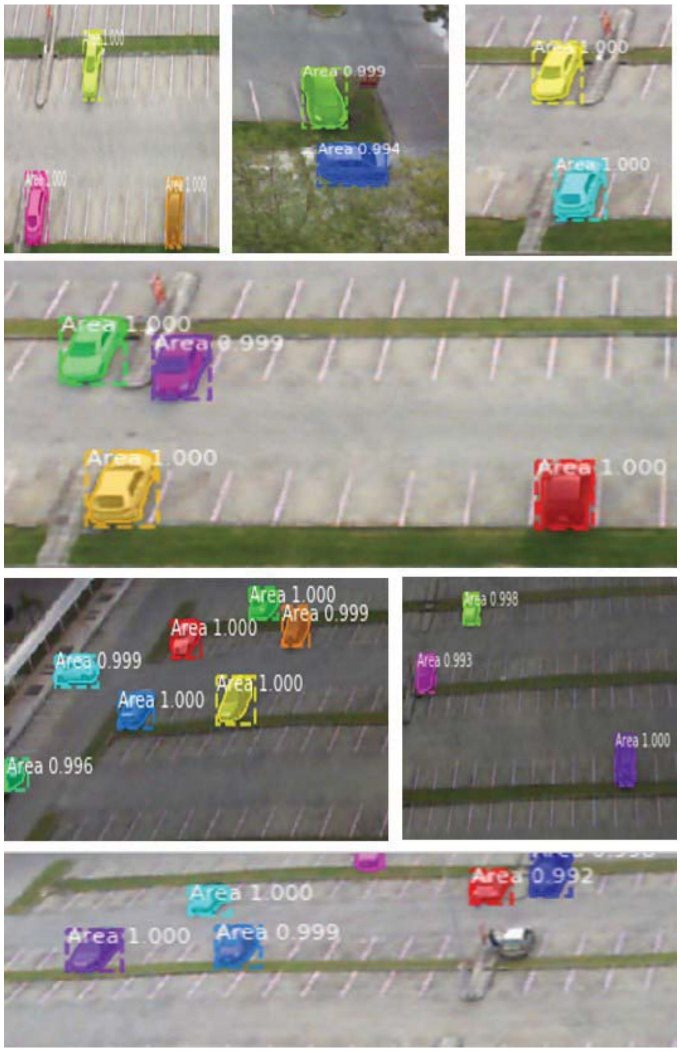

Fig. 6. Snapshots of the model's predictions

Utilizing the web cameras. the Fig.7., Fig. 8., Fig. 9., Fig. 10., Fig.11., Fig.12.shows the loss of the model alongside the training epochs. The results of the same are shown in Tables I and II.

The results from testing the vehicle classifier as shown in the Table I indicated that it is able to detect bounding boxes of the images accurately by 98.4% and masks of the vehicles accurately 92.33% of the times and with a false or wrong detection rate of 7.66%.

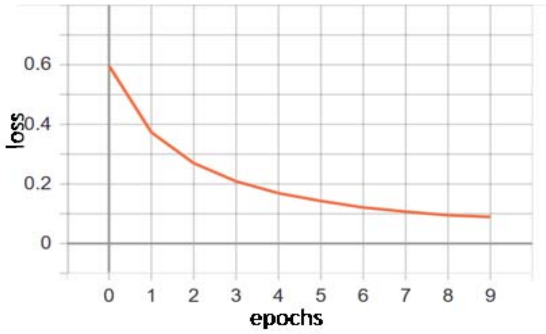

Fig. 7. Total Loss of Mask R-CNN

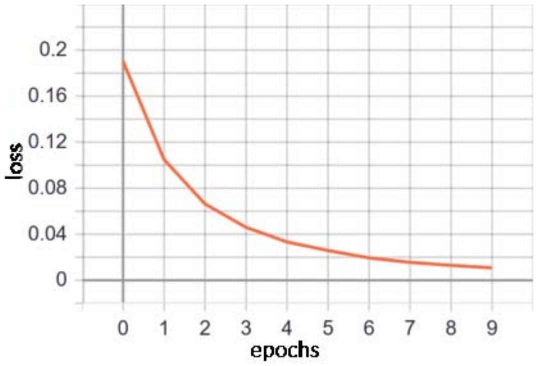

Fig. 8. Mask R-CNN bounding box loss

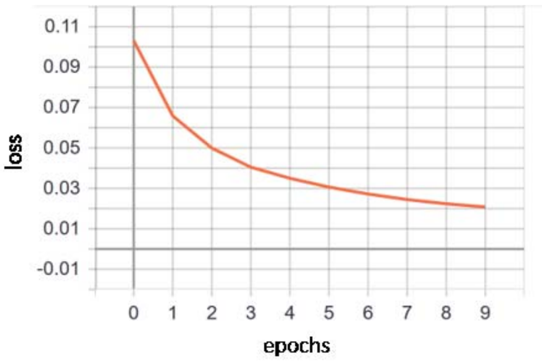

Fig. 9. Mask R-CNN class loss

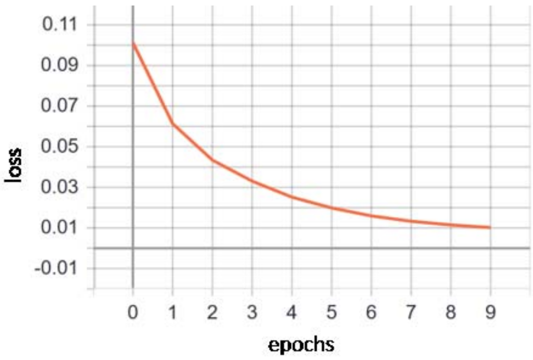

Fig. 10. Mask R-CNN mask loss

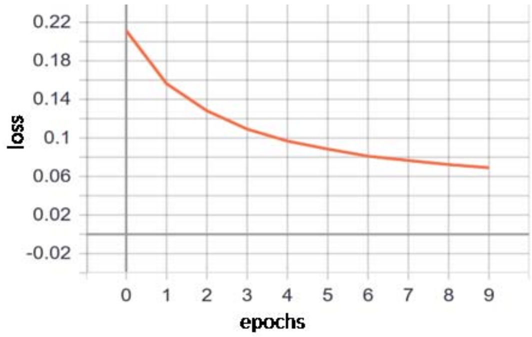

Fig. 11. RPN bounding box loss

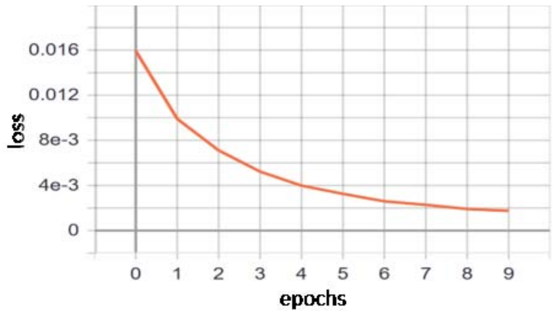

Fig. 12. RPN class loss

TABLE I
RESULTS

PKlot Dataset	*Free Slots*	*Busy Slots*	*Total Slots*
Actual number of slots	310,466	316,375	626841
Correctly identified slots	286,653	292,110	578763

TABLE II
EVALUATION METRICS

Evaluation Metrics	*Boundary Box Detection*	*Mask Detection*
Accuracy	98.4%	92.33%
Loss	1.6%	7.67%

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VII. CONCLUSION AND FUTURE SCOPE

We have implemented the method of Mask R-CNN and have got an accuracy of 92.33% for recognizing the mask of the object and an accuracy of 98.4% to detect the bounding box of the object. The method used here is just a simulation. The hardware setup can be performed in future with the help of live streaming of images, extracting the features and segmenting it manually.

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